

The Delegation Burden: A Threshold and a Separation Result for the Cost of Delegating Work to Artificial Agents

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Abstract

We define delegation cost as the human governance effort that converts an objective into an accepted outcome through an artificial agent, and we prove two results: each work unit has a unique capability threshold above which delegation beats direct execution, rising with the context specificity of the work, and delegation cost orders work units differently from transaction cost while remaining strictly positive under perfect incentive alignment, so it is neither transaction cost nor agency cost relabeled. Two pre-registered analyses of existing benchmark corpora discipline the construct's empirical reach.

1 Introduction

When Coase asked why firms exist, his answer was that using the market mechanism is costly, since “the operation of a market costs something” (Coase 392). This paper asks the same question one technological layer down: what does it cost a human to use an artificial agent, and when does that cost fall below the cost of doing the work oneself? We call the relevant quantity delegation cost, the human governance effort required to convert an objective into an acceptable outcome through an agent rather than by direct execution, and our unit of analysis is the delegable work unit, a bounded episode of work with an objective, context, constraints, an acceptance criterion, and consequences. The premise is that the capability of the agent and the delegability of the work are different quantities, because a capable agent still has to be told what to do, watched, checked, and cleaned up after when it fails. The contribution is deliberately narrow: we state the construct formally, we prove a delegability threshold result with comparative statics in context specificity, and we prove a separation result showing that delegation cost is ordered differently from transaction cost and survives the alignment that would extinguish agency cost, which answers the objection the construct most often meets, namely that it is some familiar cost in new clothing.

2 Relation to antecedents

Our claim against the nearest neighbors is one of object, not of emphasis. Transaction-cost economics after Coase and Williamson prices the friction of exchanging across a boundary between parties with interests, with governance chosen to economize on hazards of asset specificity, uncertainty, and frequency (Williamson 239); the friction priced here has no counterparty and no opportunism, since the hazard is that intent is communicated incompletely to an executor that cannot share the principal's unstated context. Principal-agent theory after Holmstrom and Milgrom derives delegation cost from divergent objectives under imperfect observability, with incentive pay distorting effort toward what is measured (Holmstrom and Milgrom 26); our agent has no objective of its own in that sense, and Proposition 2 exhibits the governance cost that remains when the alignment problem is assumed away entirely. Incomplete-contract theory after Grossman, Hart, and Moore locates the value of ownership in residual control rights when contingencies cannot be specified (Grossman and Hart 691; Hart and Moore 1120); delegation to an agent is incomplete contracting without a counterparty to own anything, so residual decision rights stay with the principal and their exercise is the cost we model rather than a right whose allocation is at issue. Team theory after Marschak and Radner, in which members share an objective but hold distributed information (Marschak and Radner 123), is the closest structural relative, and Aghion and Tirole price delegation as authority traded against initiative (Aghion

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and Tirole 1); in both, the cost of moving information and authority is the primitive we make explicit in human effort. The task-based automation literature after Acemoglu and Restrepo allocates tasks between labor and capital by comparative cost (Acemoglu and Restrepo 1495) but carries no governance term, so a task can be cheap for the machine to perform and dear for the human to delegate, which is the wedge Proposition 1 prices and capability measurement such as the METR time horizon leaves unmeasured (Kwa et al.).

Three recent contributions deserve a direct answer. Hadfield and Koh survey the economics of an economy of AI agents and call for formal apparatus they do not supply, so this paper answers their agenda rather than competing with a result. Erol, El, Suzgun, Yuksekogonul, and Zou define cost-of-pass as the expected inference cost of a correct solution against the human expert's cost, a production-cost ratio that contains no term for the principal's own specification, monitoring, or remediation effort. Huang, Xiao, and Vishnoi prove thresholds for a worker delegating under outcome-based evaluation, but their threshold lies in the worker's verification reliability and their mechanism requires misaligned evaluation, so neither the capability threshold of Proposition 1 nor the alignment-robust separation of Proposition 2 appears in their analysis.

3 The model

3.1 Primitives

A work unit is $w = (o, x, K, A, S, E, L)$: an objective, initial context, constraints, an acceptance predicate, a stakeholder and authority structure, an environment, and a loss function over failures. Three reduced-form characteristics of w carry the analysis: a direct cost $h(w) > 0$, the human effort to complete w without an agent at the required quality and risk; a context specificity $k(w) \geq 0$, the amount of principal-held information the acceptable completion of w requires; and a remediation cost $r(w) > 0$ incurred when a delegated attempt fails. Agents are indexed by capability $\theta \in [0, \infty)$. Delegation generates human governance effort

$$G(\theta; w) = s(k(w)) + m(\theta; w) + (1 - p(\theta, k(w))) r(w),$$

where s is specification effort, m is monitoring and verification effort, and $p \in [0, 1]$ is the probability the delegated attempt is acceptable. The delegation burden ratio is $\text{DBR}(\theta; w) = G(\theta; w)/h(w)$, and w is delegable at θ when $\text{DBR}(\theta; w) < 1$.

Assumption 1. $G(\cdot; w)$ is continuous and strictly decreasing in θ , with $p(\cdot, k)$ strictly increasing, $m(\cdot; w)$ nonincreasing, and s strictly increasing in k with $s(0) = 0$.

Assumption 2. $G(0; w) > h(w)$ for every w , and $G_\infty(w) := \lim_{\theta \rightarrow \infty} G(\theta; w)$ exists.

The limit in Assumption 2 exists automatically, since G is decreasing and bounded below by zero; we name it because it is the structural delegation cost, the governance effort that survives arbitrary capability.

3.2 A delegability threshold

Proposition 1. Under Assumptions 1 and 2, for every work unit w there is a unique $\theta^*(w) \in (0, \infty]$ such that $\text{DBR}(\theta; w) < 1$ if and only if $\theta > \theta^*(w)$, and $\theta^*(w)$ is finite if and only if $G_\infty(w) < h(w)$. Moreover, if $k' > k$ and $G(\theta; k') \geq G(\theta; k)$ for all θ with strict inequality at $\theta = \theta^*(k)$, which holds in particular when s is strictly increasing and p is nonincreasing in k with all else fixed, then $\theta^*(k') > \theta^*(k)$ whenever $\theta^*(k)$ is finite. The delegable set $W_D(\theta) = \{w : \theta > \theta^*(w)\}$ is nondecreasing in θ .

Proof. Fix w and write $g(\theta) = G(\theta; w) - h(w)$, which is continuous and strictly decreasing with $g(0) > 0$ by Assumption 2. If $G_\infty(w) \geq h(w)$ then $g(\theta) > g_\infty = G_\infty(w) - h(w) \geq 0$ for all finite θ , so no finite crossing exists and $\theta^*(w) = \infty$ satisfies the statement. If $G_\infty(w) < h(w)$ then $g(\theta) < 0$ for all θ large enough, so by the intermediate value theorem there is θ^* with $g(\theta^*) = 0$, and strict monotonicity makes it unique and gives $g(\theta) < 0$ exactly on $\theta > \theta^*$. For the comparative static, note $G(\theta^*(k); k') > G(\theta^*(k); k) = h$, so $g(\cdot; k')$ is still positive at $\theta^*(k)$, and since it is strictly decreasing its zero, if any, lies strictly to the right; hence $\theta^*(k') > \theta^*(k)$. Monotonicity of $W_D(\theta)$ is immediate from the threshold characterization. $\square$

The threshold formalizes the adoption claim, that work moves to agents when the risk-adjusted cost of delegating falls below the risk-adjusted cost of doing, and the comparative static says the order in which work units cross is governed by structural coordinates of the work, which is the ordering the companion program calls the intrinsic delegability index.

3.3 A separation result

Let the task space W contain, for each pair $(\sigma, k) \in [0, 1]^2$, a work unit with asset specificity σ and context specificity k , where transaction cost is $TC(w) = \tau(\sigma(w))$ for some strictly increasing τ , in the spirit of governance-choice theory. Say the executor is aligned on w if its operative objective coincides with the principal’s acceptance predicate, in which case classical agency cost, the residual loss from objective divergence under hidden action, is zero. Assume the codifiable pole is regular: at $k = 0$ the work is fully specifiable and machine-checkable, so $s(0) = 0$, $m(\theta; w) \rightarrow 0$, and $p(\theta, 0) \rightarrow 1$ as $\theta \rightarrow \infty$, hence $G_\infty(w) = 0$ when $k(w) = 0$.

Proposition 2. (i) *There exist work units $w_1, w_2 \in W$ and a finite $\bar{\theta}$ such that $TC(w_1) < TC(w_2)$ while $G(\theta; w_1) > G(\theta; w_2)$ for all $\theta > \bar{\theta}$, so the delegation-cost order is not a monotone transform of the transaction-cost order.* (ii) *There exists $w \in W$ with an aligned executor, hence zero agency cost, such that $\inf_\theta G(\theta; w) \geq s(k(w)) > 0$, so delegation cost is strictly positive where agency cost is identically zero.*

Proof. For (i) take w_1 with $(\sigma, k) = (0, 1)$ and w_2 with $(\sigma, k) = (1, 0)$. Then $TC(w_1) = \tau(0) < \tau(1) = TC(w_2)$. By the regular pole, $G(\theta; w_2) \rightarrow G_\infty(w_2) = 0$, so there is finite $\bar{\theta}$ with $G(\theta; w_2) < s(1)$ for all $\theta > \bar{\theta}$, while $G(\theta; w_1) \geq s(k(w_1)) = s(1)$ for every θ because the specification term does not depend on θ and the remaining terms are nonnegative. Hence $G(\theta; w_1) \geq s(1) > G(\theta; w_2)$ on $\theta > \bar{\theta}$. For (ii) take $w = w_1$ with an aligned executor: agency cost is zero by alignment, while $G(\theta; w) \geq s(1) > 0$ for all θ by the same bound, and $s(1) > s(0) = 0$ by strict monotonicity of s . $\square$

The economic content of part (i) is that the frictions price different things, transaction cost rising with relationship-specific investment and the hazards of exchange while delegation cost rises with how much of the principal’s private context the work consumes, so a work unit can be cheap on one dimension and dear on the other. Part (ii) answers the reduction to agency theory: an agent with no interests of its own, perfectly obedient to its instruction, still costs $s(k)$ to instruct, because alignment with a stated objective is not possession of the unstated context that completes the objective. The wedge is specification, not motivation.

4 Empirical implications

Two pre-registered companion analyses of existing benchmark corpora discipline the construct’s empirical reach, and both are reported net of degenerate baselines, a precaution the benchmarks themselves make necessary, since a do-nothing agent passes 38 percent of the airline tasks of a leading tool-agent benchmark because unsolvable tasks score as passed when the environment is unchanged (Zhu et al.). On the public leaderboard corpus of that benchmark’s successor, fifteen frontier submissions with four trials per task (Yao et al.), a burden proxy built from cross-trial flakiness and interaction effort ordered tasks differently from headline failure at the top of the capability ladder, with a median rank correlation of 0.58 against a registered bar of 0.7, but the registered persistence hypothesis was falsified, since the divergence narrows significantly up the ladder, and the registered robustness check shows the narrowing is carried by the flakiness component, which is mechanically coupled to success, and reverses without it. On 1,302 expert-annotated web-agent trajectories (Lù et al.), the registered null applies: the proxy predicted the human side-effect label at 0.58 against a 0.70 bar, and predicted sub-optimality at 0.89 but only 0.009 better than trajectory length alone. The pattern is consistent with the model’s reading that signals observable in agent transcripts track the reliability term $(1 - p)r$, which capability erodes, while the specification term $s(k)$, which Proposition 2 locates as the surviving wedge, is invisible to them, though we registered no test of that interpretation and offer it as interpretation only.

5 Limitations

The theory prices human governance effort, but none of its central ratios can be measured without human data, since the measurand of the delegation burden ratio and its relatives is human time and attention, and the companion analyses observe only agent-side signals; a predictor validated against such signals, or against existing human annotations of trajectories, is a predictor and not a measurement, and any number reported as a burden ratio from such data assumes the very terms a human-subject study would measure. The reduced forms are correspondingly coarse, since s , m , and p summarize what a richer theory would derive, the regular-pole assumption locates full verifiability at zero context specificity by construction, and the separation result is an existence claim over a rich task space rather than a measurement of where actual work units sit. The results establish that the construct is coherent and distinct; how large its terms are is an empirical question this paper does not answer.

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